D07 · Cross-Docking & Flow-Through

01 · Cross-docking design: direct, indirect & opportunistic

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1 WHAT IT IS

Cross-docking moves received product straight from the inbound dock to outbound shipping with little or no storage. Three models exist: direct (supplier pre-sorts by destination), indirect (the DC breaks and re-sorts loads) and opportunistic (the WMS diverts a receipt that matches an open order).

2 WHY IT MATTERS

Cross-docking removes the two most expensive warehouse activities, put-away and picking, and drives inventory at the node toward zero **(Apte and Viswanathan, 2000 — International Journal of Logistics: Research and Applications)**. **But it only exploits synchronization that already exists between supply and demand; it does not create it.** Applied to unreliable suppliers or erratic demand, it turns a storage problem into dock congestion.

3 HOW IT IS DONE

1. **Select SKUs by rotation and supplier reliability.** High, stable demand plus predictable delivery windows.
2. **Start opportunistic.** WMS logic that diverts matching receipts needs nothing from suppliers.
3. **Build supplier capability for direct cross-dock.** SSCC labels, accurate ASNs and delivery windows, agreed and measured.
4. **Measure dwell time honestly.** Hours from receipt to dispatch; a day on the dock is storage without racks.

IN PRACTICE

Select SKUs for cross-docking on supplier reliability — not just rotation

High rotation makes a SKU attractive; supplier lead-time reliability makes it feasible. A high-rotation SKU from a supplier with variable delivery windows will congest the dock rather than flow through it.

Start with opportunistic cross-docking — it requires nothing from suppliers

Direct cross-docking needs supplier pre-sorting and labeling, which takes months to negotiate. Opportunistic cross-docking is WMS logic alone: when a receipt matches an open order, divert it to dispatch. It is the fastest route to a first result.

Direct cross-docking is a vendor compliance program in disguise

It depends entirely on the supplier shipping pre-sorted, correctly labeled, accurately quantified loads. Without the SSCC and ASN standards behind it, direct cross-docking fails at the receiving dock.

Do not cross-dock a SKU whose demand is not already synchronized with its supply

Cross-docking does not create synchronization — it exploits synchronization that already exists. Applied to unsynchronized flows, it converts a storage problem into a dock congestion problem, which is worse because dock space is scarcer than rack space.

ILLUSTRATIVE CASE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case: Cross-dock program design — distribution DC, 4,200 active SKUs — The DC stores and picks every SKU. Analysis of rotation, supplier reliability, and shelf life to identify cross-dock candidates.

SKU segment	Model selected	Rationale
High-rotation SKUs from 3 pre-sorting suppliers	Direct cross-dock	Supplier ships pre-labeled by destination · Put-away and picking eliminated entirely

SKU segment	Model selected	Rationale
High-rotation SKUs from suppliers without pre-sort capability	Indirect (break-bulk)	Requires sorting zone · Still removes storage and picking
Mid-rotation SKUs with open orders at receipt	Opportunistic	WMS diverts to dispatch when a receipt matches an open order · No supplier change
Low-rotation, long shelf life, variable lead time	Conventional storage	Forcing cross-dock would congest docks without reducing cost

Result (illustrative): Roughly a quarter of volume qualified for some cross-dock model. The decisive constraint was not warehouse capability but supplier lead-time reliability — direct cross-dock was viable only for the three suppliers whose delivery windows were predictable enough to synchronize against.

HOW IT IS MEASURED

Cross-Dock Rate % (volume moved without put-away)

Cross-Dock Rate %(Units or pallets dispatched without a storage put-away transaction / Total units or pallets received) × 100

Benchmark: Set by the SKU profile, not a universal target · Track the trend against the volume you qualified as cross-dock eligible A cross-dock rate below the eligible volume means qualified SKUs are still being stored — usually because inbound and outbound windows are not synchronized, not because the SKU was wrongly selected.

Cross-Dock Dwell Time (time from inbound receipt to outbound dispatch)

Cross-Dock Dwell TimeAverage time from receipt confirmation to dispatch confirmation for cross-docked units

Benchmark: Hours, not days · Dwell time approaching a full day means the product is being stored, whatever the process is called Dwell time is the honest test of whether cross-docking is real. Product sitting on the dock for 24 hours is storage without racks — it carries the cost of both models and the benefit of neither.

WHAT YOU WOULD USE

Cross-Dock Execution

SAP EWM — Cross-Docking — Module: Planned + Opportunistic Cross-DockingSAP EWM supports planned cross-docking (pre-allocated at receipt) and opportunistic cross-docking, where the system detects a match between an inbound receipt and an open order and diverts the product to dispatch.

Manhattan Associates WMS — Flow-Through — Module: Cross-Dock Allocation + Dock CoordinationManhattan cross-dock module with inbound-to-outbound allocation logic and coordination against dispatch windows.

4 SO WHAT

Flow what is synchronized, store what is not. Select by rotation and supplier reliability, start with opportunistic logic, build supplier compliance for direct flows and watch dwell time.

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